

Reproduction and Analysis of RISE: Randomized Input Sampling for Explanation of Black-box Models

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1 Introduction

Deep neural networks achieve strong performance in many domains, but their predictions are often difficult to interpret. Understanding which parts of the input influence a model’s decision is therefore an important problem.

RISE (Randomized Input Sampling for Explanation of Black-box Models) is a perturbation-based explanation method that estimates feature importance by randomly masking parts of the input and measuring the effect on the model prediction. Because it only requires model outputs, RISE can be applied in a fully black-box setting.

In this project, we reproduce the RISE method from scratch and evaluate it on image classification using a pretrained ResNet-50 model on the Imagenette dataset. We compare RISE with Grad-CAM using qualitative visualizations and quantitative metrics such as deletion and insertion curves. We also analyze the impact of the number of masks on the stability of the explanations.

Finally, we extend the analysis to a different modality: protein sequences. Using an ESM2 model fine-tuned for Pfam classification, we apply RISE to identify which amino acids contribute to functional domain predictions, and study how explanations behave in a multi-label setting.

2 Related Work and State of the Art

2.1 Gradient-based methods

A large class of interpretability techniques relies on gradients of the model output with respect to the input. Early approaches such as Saliency Maps compute the gradient of the class score with respect to the input pixels to estimate their importance. More advanced methods attempt to improve the stability or interpretability of these gradients. For instance, Integrated Gradients accumulates gradients along a path between a baseline input and the actual input, providing a theoretically grounded attribution method. Grad-CAM, one of the most widely used visualization techniques for convolutional neural networks, produces class-discriminative localization maps by combining feature map activations with gradient information from the final convolutional layers. While these methods are computationally efficient, they require access to the model gradients and therefore operate in a white-box setting.

2.2 Perturbation-based methods

Another family of approaches evaluates feature importance by observing how model predictions change when parts of the input are modified or removed. Occlusion-based methods, for example, systematically mask regions of the input and measure the resulting change in prediction confidence.

Although conceptually simple and model-agnostic, exhaustive occlusion can be computationally expensive because it requires a large number of forward passes through the model.

RISE belongs to this category of perturbation-based explanations. Instead of deterministically masking individual regions, RISE samples random masks and estimates feature importance through Monte Carlo averaging. This stochastic formulation makes it possible to approximate pixel importance efficiently while maintaining a purely black-box setting, meaning that only model outputs are required.

2.3 Game-theoretic attribution methods

A third line of work is based on concepts from cooperative game theory. Methods such as SHAP approximate Shapley values, which provide theoretically justified feature attributions satisfying several desirable properties, including additivity and fairness. However, computing exact Shapley values is generally intractable for high-dimensional inputs, and practical implementations rely on approximations that can still be computationally expensive.

3 Method: RISE

3.1 Problem Formulation

Given a trained classifier f and an input image $I \in \mathbb{R}^{H \times W \times C}$, our goal is to compute a saliency map $S \in \mathbb{R}^{H \times W}$ that reflects the importance of each pixel for a target class c . The saliency value at each pixel should quantify how much that pixel contributes to the model prediction $f_c(I)$.

3.2 Random Mask Sampling

RISE estimates pixel importance by probing the model with randomly masked versions of the input. We generate N random binary masks $\{M_i\}_{i=1}^N$, sampled at low resolution and upsampled to the input size. Each mask is drawn independently, with each pixel activated with probability p_1 . For each mask M_i , we create a masked image:

$$I_i = I \odot M_i$$

where $\odot$ denotes element-wise multiplication.

3.3 Monte Carlo Estimation

The saliency map is computed using a Monte Carlo approximation:

$$S = \frac{1}{N} \sum_{i=1}^N f_c(I \odot M_i) M_i$$

where:

- $f_c(I \odot M_i)$ is the confidence score for class c ,
- M_i indicates which pixels were visible.

Intuitively, pixels that frequently appear in masks yielding high confidence will receive higher saliency values. As N increases, the estimator becomes more stable.

4 Evaluation Metrics

To assess the faithfulness of saliency maps, we use deletion and insertion metrics.

4.1 Deletion Metric

Pixels are removed in decreasing order of saliency. At each step t , we measure the model confidence $f_c(I_t)$ after removing the top- t fraction of pixels. A good saliency method should cause the confidence to drop rapidly. We report the Area Under the Curve (AUC). Lower AUC indicates better faithfulness.

4.2 Insertion Metric

Starting from a blurred (or baseline) image, pixels are progressively revealed according to their saliency ranking. A good saliency method should lead to a rapid increase in confidence. We report the AUC of the insertion curve. Higher AUC indicates better performance.

5 Datasets and Models

To evaluate the generality of our approach, we conduct experiments on two different data modalities: natural images and amino acid sequences.

5.1 Datasets

For the image modality, we use Imagenette2-320, a curated subset of ImageNet containing 10 object classes. All images are resized to 224×224 , which corresponds to the standard input resolution for ResNet architectures. We apply the standard ImageNet normalization (mean and standard deviation) before feeding the images to the model.

To explore a different modality, we also evaluate the method on protein sequences. We use the SwissProt-Pfam dataset, which consists of amino acid sequences annotated with Pfam functional labels. The task is formulated as a multi-label classification problem, where each sequence may belong to multiple Pfam families.

5.2 Models

For image classification, we use a ResNet-50 model pre-trained on ImageNet-1K. The model is used without additional fine-tuning, allowing us to evaluate explanation methods directly on a standard pretrained classifier.

For protein sequences, we use an ESM2 transformer model fine-tuned for Pfam classification. The model is loaded using AutoTokenizer and AutoModelForSequenceClassification from the HuggingFace Transformers library.

This setup allows us to evaluate RISE explanations not only on images but also on biological sequences, highlighting the ability of the method to generalize across modalities.

6 Results on images modality

6.1 Results on examples

We first compare RISE and Grad-CAM on several illustrative examples. On the selected image, where the model predicts the class French horn, we observe a clear qualitative difference between the two methods.

RISE produces saliency maps that are more tightly localized on the object itself. The most important pixels are concentrated around the instrument, leading to a sharper and more object-focused attribution. In contrast, Grad-CAM tends to generate more diffuse heatmaps. The highlighted regions extend beyond the object boundaries, covering surrounding areas that are less semantically relevant to the prediction.

This behavior suggests that RISE provides finer spatial localization, while Grad-CAM captures broader, coarser regions due to its reliance on feature map activations at deeper convolutional layers.

Additional qualitative examples are provided in the appendix to support the consistency of this observation across different images and predicted classes.

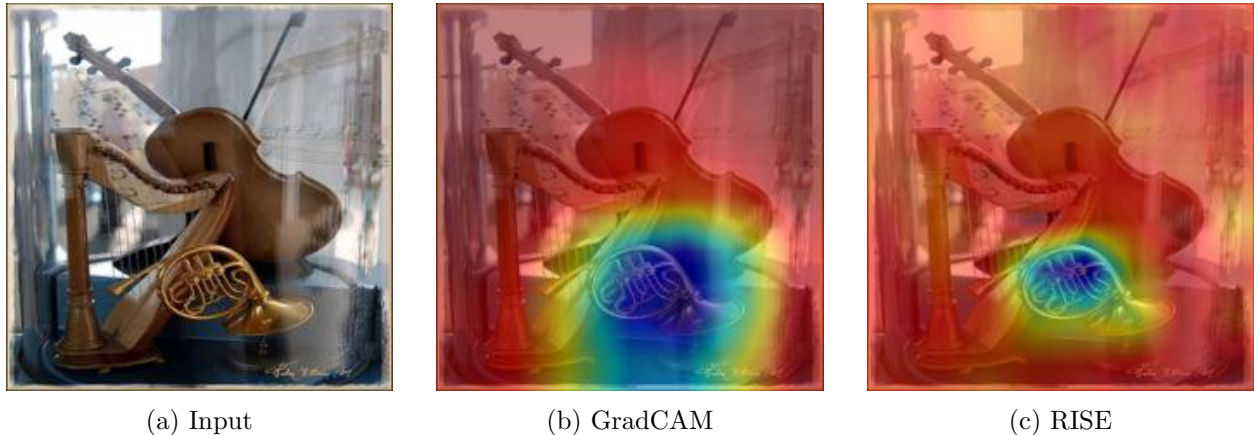


Figure 1: Comparison between the input image, GradCAM and RISE saliency maps.

To complement the visual analysis, we also compare the two methods using quantitative evaluation metrics, namely deletion and insertion curves. These metrics allow us to assess how quickly the model confidence drops when the most important pixels are removed (deletion), and how efficiently the confidence increases when important pixels are progressively introduced (insertion). By plotting these curves for both Grad-CAM and RISE, we obtain a more objective comparison of their attribution quality.

For the deletion metric, the model confidence decreases a little bit faster when removing the most important pixels identified by RISE. This indicates that RISE more accurately captures the truly discriminative regions: once these pixels are removed, the model quickly loses evidence for its prediction. In contrast, the confidence drops more gradually with Grad-CAM, suggesting that its highlighted regions contain more irrelevant or less critical pixels.

For the insertion metric, the confidence increases more rapidly when progressively adding the pixels ranked by RISE. This means that the regions identified by RISE provide stronger and more concentrated evidence for the predicted class. Grad-CAM, on the other hand, requires more pixels to be inserted before reaching a similar confidence level, reflecting less precise localization.

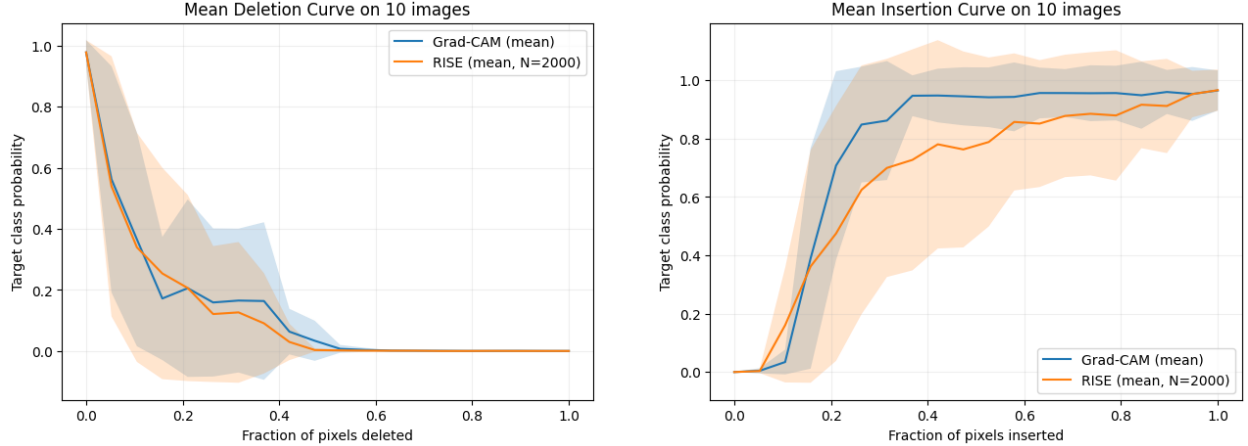


Figure 2: Deletion and Insertion curves for few images

6.2 Importance of the Number of Masks

The number of masks N is a critical hyperparameter in RISE, as the method relies on a Monte Carlo estimation of pixel importance. The saliency map is computed as

$$S = \frac{1}{N} \sum_{i=1}^N f_c(I \odot M_i) M_i, \quad (1)$$

which can be interpreted as an empirical average of i.i.d. random variables. Under this formulation, RISE can be seen as an unbiased estimator of the expected contribution of each pixel under random masking. However, as with any Monte Carlo estimator, its variance decreases at rate $\mathcal{O}(1/N)$. Consequently, for small values of N , the estimator exhibits high variance, leading to unstable and noisy saliency maps.

To quantify this effect, we measured the correlation between saliency maps computed with varying numbers of masks and a reference map obtained using 5000 masks. The similarity increases monotonically with N , confirming convergence toward a stable estimate. For very small values (e.g., 200 masks), the correlation remains below 0.5, indicating that the explanation is still largely dominated by sampling noise. This empirically illustrates the high-variance regime predicted by Monte Carlo theory.

We further observe that computational cost scales approximately linearly with N , since each mask requires a forward pass through the model. This creates a direct trade-off between explanation stability and runtime. In addition, we evaluated faithfulness metrics (Deletion AUC and Insertion AUC) as functions of N . For small numbers of masks, the curves are highly noisy and unstable across runs. As N increases, the metrics progressively improve: Deletion AUC decreases (indicating that removing the most salient pixels rapidly decreases model confidence), while Insertion AUC increases.

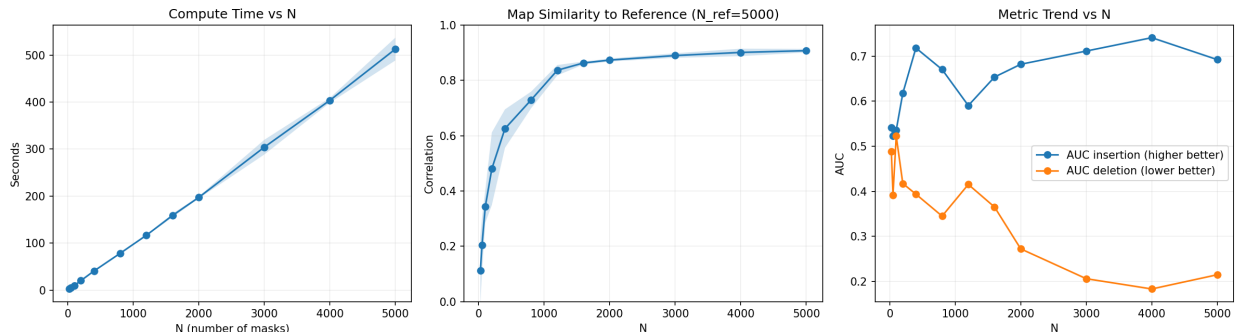


Figure 3: Number of masks sensitivity

All experiments were repeated multiple times with different random seeds, and the plots display the associated variance. These results confirm that insufficient sampling leads not only to visually unstable explanations but also to unreliable faithfulness estimates, whereas larger N stabilizes both qualitative and quantitative behavior. Importantly, this highlights that RISE’s main limitation is not bias but variance, which is fundamentally controlled by the sampling budget.

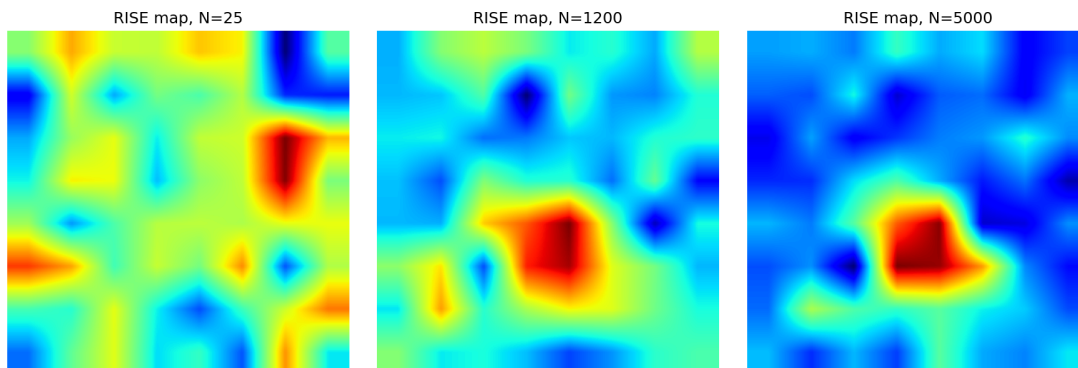


Figure 4: RISE maps

We can also visualize the RISE maps for small, mid, and large number of mask, we observe that we have much more homogenous results for a large number of masks.

7 Results on the amino acid sequence modality

To evaluate the generality of RISE beyond images, we also apply the method to the amino acid sequence modality. In this setting, the input is a protein sequence, which is a chain of amino acids.

The sequence is first tokenized, where each position corresponds to an amino acid (or a sub-token depending on the tokenizer). The tokenized sequence is then fed into an ESM2 transformer model that has been fine-tuned for Pfam classification. The model outputs a score for each Pfam class. The task is formulated as a multi-label classification problem, since a single protein may contain several functional domains and therefore belong to multiple Pfam families simultaneously.

Pfam is a large database of protein families and domains. Each Pfam entry corresponds to a conserved functional domain or motif commonly observed across proteins. Predicting Pfam labels therefore amounts to identifying which functional domains are present in a given protein sequence.

Once the model produces its predictions, we apply RISE to compute a saliency map over the sequence positions. This produces an importance score for each amino acid, indicating which positions in the sequence contribute the most to the model’s prediction of specific Pfam domains. Such explanations can help highlight biologically relevant regions of the protein that drive the classification.

7.1 Example

We consider the following protein sequence:

MSKIIIEYDETARRAIEAGVNTLADAVRVTLGPRGRHVVLAKAFGGPAVTNDGVTVAREID...

which contains 540 amino acids. The goal is to identify which amino acids are responsible for the strong prediction of the Pfam family 54, among the 1000 possible Pfam classes predicted by the model.

To analyze this, we apply the RISE explanation method by randomly masking amino acids in the sequence. In this context, masked positions are replaced with the special token <MASK>. By observing how the prediction score changes when different positions are masked, we can estimate the importance of each amino acid for the prediction.

The resulting importance map is shown in Figure 5.

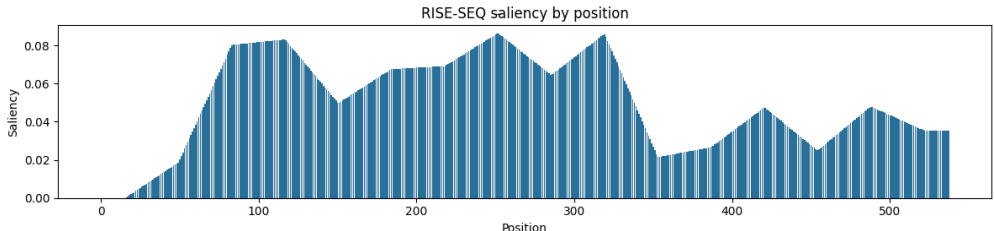


Figure 5: RISE importance scores for each amino acid position with respect to the prediction of Pfam 54.

The x-axis corresponds to amino acid positions along the sequence, while the y-axis represents their importance in predicting the Pfam 54 label. We observe that many amino acids contribute to the prediction rather than only a small subset. In contrast to image classification tasks, where explanations are often highly localized, the importance here is more distributed across the sequence. Apart from the first few amino acids, most positions appear to contribute some information to the prediction.

To further evaluate the explanations, we also examine the deletion and insertion curves.

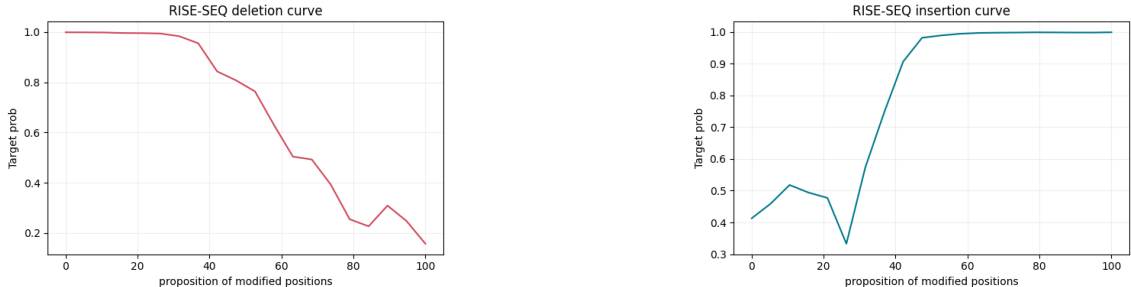


Figure 6: Deletion and insertion curves for the RISE explanation.

The results are particularly interesting. When at least 40% of the amino acids are present, the probability of predicting Pfam 54 is already close to 1. This suggests that only a fraction of the sequence is sufficient for the model to make a confident prediction.

Even more strikingly, when less than 40% of the amino acids are present, the model can still predict Pfam 54 with relatively high probability. In the insertion setting, the probability exceeds 0.5 with only 15% of the amino acids included.

A key aspect explaining this behavior is that the predicted class is not unique. Protein sequences typically belong to multiple Pfam families. In our example, the model predicts 104 Pfam families (out of 1000) with a probability greater than 0.9 for this sequence. Therefore, the problem differs significantly from datasets such as Imagenette, where each image belongs to a single class.

7.2 RISE Analysis Across Multiple Pfam Predictions

For a given protein sequence, the model may predict several Pfam families with high probability. Rather than explaining a single prediction, it is therefore informative to analyze how different amino acids contribute to multiple Pfam predictions simultaneously.

To investigate this, we extend the RISE analysis by applying it independently to several Pfam classes for the same amino acid sequence. This allows us to identify:

- which amino acids contribute to a specific Pfam prediction,
- which residues influence multiple Pfam families simultaneously.

We focus on the top-10 Pfam predictions produced by the model for the studied sequence. Table 1 reports the Pfam indices together with their predicted probabilities.

Pfam Index	Predicted Probability
54	0.9989
327	0.9966
288	0.9946
167	0.9900
118	0.9893
610	0.9877
583	0.9865
643	0.9842
198	0.9813
368	0.9798

Table 1: Top-10 Pfam predictions for the analyzed protein sequence.

We then apply RISE to each of these Pfam classes independently. The resulting saliency maps highlight the amino acid positions that most influence the prediction of each Pfam family.

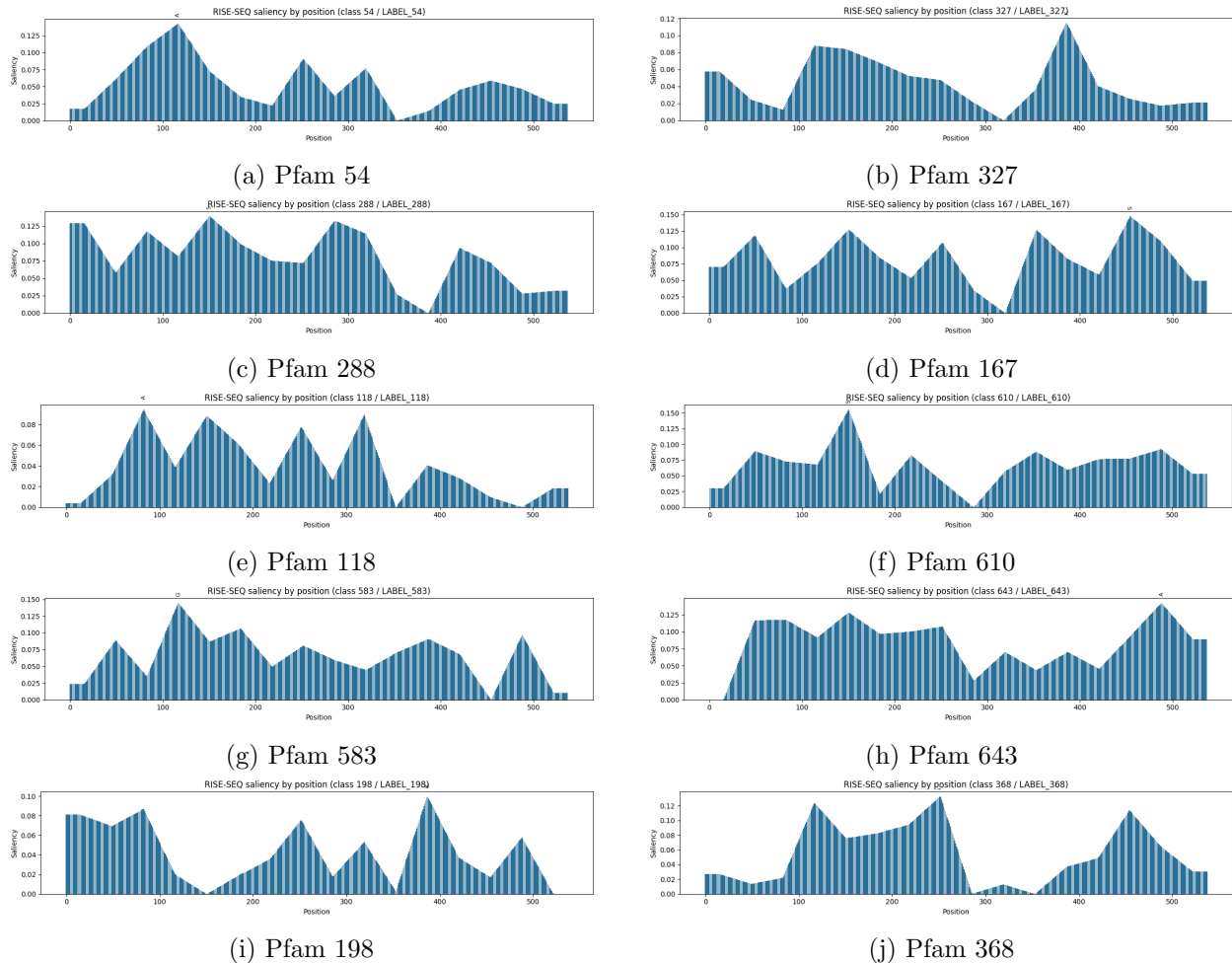


Figure 7: RISE saliency maps computed for the top-10 Pfam predictions of the sequence. Each map highlights the amino acid positions that contribute most to the corresponding Pfam classification.

From these saliency maps, we observe that almost every amino acid appears to contribute to the prediction of at least one Pfam family. To better visualize this global contribution, we compute the average saliency across the ten RISE maps corresponding to the top-10 Pfam predictions.

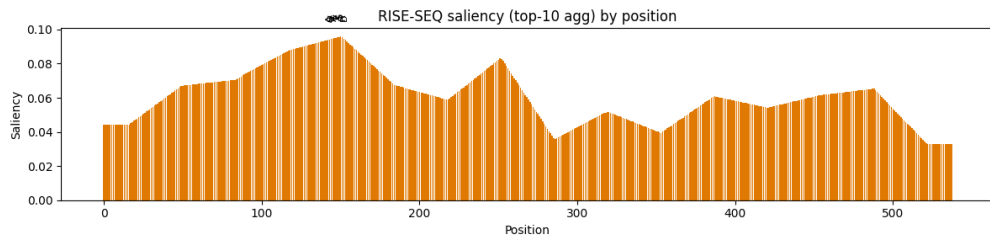


Figure 8: Average RISE saliency map over the top-10 predicted Pfam families.

The resulting averaged map highlights that most amino acid positions contribute to the model's predictions. In contrast to image classification tasks, where discriminative information is often spatially localized, the informative signal in protein sequences appears to be more globally distributed.

Furthermore, in the multi-label setting considered here, a single amino acid can influence the prediction of several Pfam families simultaneously. This differs from typical image classification scenarios, where a pixel belonging to a specific object usually contributes primarily to the prediction of that object and rarely to others.

8 Use of LLMs

During this project, we leveraged large language models (LLMs) and code-focused assistants to improve both code quality and report clarity. We used AI tools such as Codex to help organize, clean, and standardize our code. This included adding type hints, ensuring adherence to conventional Python norms, and refactoring functions for readability and maintainability. Such practices improved code readability and facilitated debugging and reproducibility.

LLMs were also used to proofread and edit the report, ensuring proper grammar, style consistency, and clarity of explanations. Additionally, the models helped with translations between French and English.

9 Conclusion

In this project, we reproduced the RISE explanation method and evaluated its behavior on different modalities. On image classification tasks, RISE produces meaningful saliency maps and performs competitively with Grad-CAM according to deletion and insertion metrics.

We also showed that the number of sampled masks strongly influences the stability of the explanations, highlighting the variance-computation trade-off inherent to Monte Carlo estimation. Finally, by applying RISE to protein sequences, we demonstrated that the method can be extended beyond images. In this setting, explanations tend to be more globally distributed across the sequence, and individual amino acids may contribute to several Pfam predictions due to the multi-label nature of the task.

Overall, RISE provides a simple and model-agnostic framework for interpreting black-box models across different data modalities.